MILITARY SPECIFICATION

CARTRIDGE, 4.2 INCH, CS SMOKE, XM630 LOADING, ASSEMBLING, AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling, and packing of smoke cartridge XM630 for 4.2 inch mortars.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent stated herein.

SPECIFICATIONS

MILITARY

MIL–A–2550 — Ammunition and Special Weapons, General Specification for.

MIL–I–45607 — Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes (ABC–STD–105).

MIL–STD–109 — Quality Assurance Terms and Definitions.

MIL–STD–1168 — Lot Numbering of Ammunition.

MIL–STD–1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ARMY

9220299 — Cartridge, 4.2 Inch, XM630 Assembly.

7549247 — Container, Ammunition, Fiber, M251 for 4.2 Inch Mortar Ammo, Assembly, Details, and Packing.

7549248 — Box, Packing, Ammunition For 4.2 Inch Mortar Ammo, in Fiber Container M251 Assem., Details and Packing.

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PUBLICATIONS

ARMY

ET–9220299 — Equipment Tabulation.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Material and parts shall be in accordance with applicable drawings and specifications.

3.2 Cartridges. Cartridges shall comply with all requirements specified on Drawing (dwg.) 9220299 and with all requirements specified in applicable specifications.

3.3 Proving ground. The cartridge shall comply with the following requirements when tested as specified in 4.3.1.

3.3.1 *Functioning.* The projectile shall function satisfactorily (see 6.4) and shall burn for a minimum time of 30 seconds.

3.3.2 *Premature bursts and security of parts.* There shall be no premature burst or indication of metal parts failure in the gun bore or in the first 200 yards of horizontal range.

3.4 Workmanship. All manufacturing, processing and assembly operations of the cartridge shall be correctly performed in a thoroughly workmanlike manner. Parts shall be free of burrs, chips, dirt, cracks, grease, rust, and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL–STD–109 in order to define the terms used herein. The provisions of MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspections performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications refer-

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enced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings, and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL–STD–105 and shall be numbered in accordance with MIL–STD–1168. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots for components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Each lot shall contain:

(a) Metal parts from one lot interfix number from one manufacturer.

(b) Ejection charges of one lot interfix number from one manufacturer.

(c) Canisters loaded at one plant using the same formula and process under the same drawing and revision, and the same specification and revision.

(d) Propelling charges from not more than one lot.

(e) Ignition cartridges of one lot interfix number from one manufacturer.

(f) Loaded fuzes of one lot interfix number from one manufacturer.

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(g) Obturating mechanisms of one lot interfix number from one manufacturer.

4.2.2 *Examination.* Inspection for Critical defects (and Major defects, when so specified) shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL–STD–105 except that continuous sampling plans in accordance with MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Projectile, prior to assembly of expelling charge (see dwg. 9220298 covering a detail of dwg. 9220299).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Foreign matter in cavity	Visual	01001
Minor:	None defined.		

4.2.2.2 Projectile, prior to assembly of base plug and obturating mechanism (see dwg. 9220298 covering a detail of dwg. 9220299).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Any component missing or improperly assembled	Visual	02001
Minor:	None defined.		

4.2.2.3 Projectile prior to assembly of fuze (see dwg. 9220298 covering a detail of dwg. 9220299).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Gap between base plug and body excessive	Gage	03001
102.	Weight zone incorrect	Balance	03002

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Categories	*Defects*	*Method of Inspection*	*Code No.*
Minor:	AQL 1.5 percent		
201.	Shear or twist pin above flush	Visual	03003
202.	Pins loose or stakes missing	Visual-Manual	03004
203.	Marking missing, misleading or unidentifiable	Visual	03005
204.	Evidence of poor workmanship (see 3.4)	Visual	03006

4.2.2.4 Cartridge assemby, prior to assembling pressure plate (see dwg. 9220299).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Cartridge container fails to withstand dissassembly torque	Test	04001
102.	Rotating disc not properly assembled	Visual	04002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	04003

4.2.2.5 Cartridge assembly after assembly of pressure plate and pressure plate nut (see dwg. 9220299).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Pressure plate assembled backwards	Visual	05001
102.	Nut improperly staked	Visual	05002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	05003

4.2.2.6 Cartridge assembly after assembling ignition cartridge and striker nut assembly (see dwg. 9220299).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Striker nut fails to withstand disassembly torque	Test	06001
102.	Ignition cartridge missing	Visual	06002
103.	Striker point missing or loose	Visual-Manual	06003

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Categories	*Defects*	*Method of Inspection*	*Code No.*
Minor:	None defined.		

4.2.2.7 Cartridge assembly after assembling extension and pin (see dwg. 9220299).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Pin missing	Visual	07001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	07002

4.2.2.8 Cartridge assembly after assembling propelling charge (see dwg. 9220299 and 8863617 covering a detail of dwg. 9220299).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:			
1.	Excessive number of increments	Visual	08001
2.	Warning label missing	Visual	08002
Major:	AQL 0.65 percent		
101.	Insufficient number of increments	Visual	08003
102.	Bag loading assembly missing	Visual	08004
103.	Any propellant holder missing	Visual	08005
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	08006

4.2.2.9 Cartridge assembly, prior to assembling Fuze (see dwg. 9220299).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Expelling charge missing	Visual	09001
102.	Adapter missing or damaged	Visual	09002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	09003

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4.2.2.10 Cartridge assembly, prior to packing (see dwg. 9220299).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Fuze or adapter fails to withstand disassembly torque	Test	10001
102.	Any component damaged or improperly assembled	Visual-Manual	10002
Minor:	AQL 1.5 percent		
201.	Fuze not staked	Visual	10003
202.	Marking misleading or unidentifiable including incorrect color code	Visual	10004
203.	Protective coating with an aggregate of bare areas in excess of ¼ square inch	Visual	10005
204.	Evidence of poor workmanship (see 3.4)	Visual	10006

4.2.2.11 Unsealed fiber container (see dwg. 7549247).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Spacer tube missing	Visual	11001
102.	Support tube missing	Visual	11002
Minor:	AQL 0.65 percent		
201.	Round cannot be removed by hand	Manual	11003

4.2.2.12 Sealed fiber container (see dwg. 7549247).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Sealing tape incomplete or badly wrinkled	Visual	12001
102.	Metal end loose or distorted	Visual-Manual	12002
103.	Container damaged or cut through all asphalt-impregnated layers	Visual	12003
Minor:	AQL 2.5 percent		
201.	Gap between body and cover of container exceeding ⅛ inch	Visual	12004

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Categories	Defects	Method of Inspection	Code No.
202.	Contents loose	Manual	12005
203.	Tear tab length inadequate	Visual	12006
204.	Marking misleading or unidentifiable	Visual	12007
205.	Evidence of poor workmanship (see 3.4)	Visual	12008

4.2.2.13 Sealed packing box (see dwg. 7549248).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Box damaged exposing contents	Visual	13001
102.	Hardware or strapping missing or loose	Visual-Manual	13002
Minor:	AQL 1.0 percent		
201.	Car seal improperly installed	Visual	13003
202.	Marking misleading or unidentifiable	Visual-Manual	13004

4.2.3 *Testing.*

4.2.3.1 *Deterioration check.* (See applicable ignition cartridge specification) Major defect — Code No. 05004. If the time between the original acceptance of an ignition-cartridge lot and the assembly of that lot into the complete-round cartridges exceeds two years, or if the ignition-cartridges have been exposed to adverse conditions, however brief, at any time since previous tests, the ignition cartridge lot shall be subjected to and must satisfactorily pass the check test specified in the applicable ignition cartridge loading specification, immediately before the ignition cartridges are assembled into complete round cartridges. The test shall be performed by the contractor on ignition cartridges selected by the Government inspector at the facility assembling the ignition cartridges into the complete-round cartridges (see 6.7).

4.2.3.2 *Profile and alignment* (see dwg. 9220299) Major defect — Code No. 10007. The cartridges shall be gaged 100 percent for profile and alignment. Any cartridge which fails to meet the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.2.4.

4.2.3.3 *Proving ground.* The following tests shall be performed as specified in 4.3.1.

4.2.3.3.1 *Functioning* (see 3.3.1).

4.2.3.3.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, forty eight sample cartridges shall be selected for this test. The cartridges shall be fired in three groups of sixteen rounds each at propelling charge levels of 12, 25½ and 41 increments. The cartridges shall be observed for compliance with the criteria specified in table 1. Testing shall be in accordance with 4.3.1.

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4.2.3.3.1.2 After three consecutive lots have complied with the criteria of 4.2.3.3.1.1, fifteen sample cartridges shall be selected for this test. The cartridges shall be fired in three groups of five rounds each at propelling charge levels of 12, 25½ and 41 increments. The cartridges shall be observed for compliance with the criteria specified in table I. Testing shall be in accordance with 4.3.1.

TABLE I.

Category	*Defect*	*Acceptance Criteria*				*Code No.*
	Sample Size	*48 (see 4.2.3.3.1.1)*			*15 (see 4.2.3.3.1.2)*	
		accept	*reject*	*accept*	*reject*	
Major	Failure to fire	0	1	0	1	10008
Major	Functioning	2	3	1	2	10009
Major	Projectile provides smoke for less than 30 seconds	3	4	1	2	10010

Note. If more than 4 rounds in a lot have a major defect, the lot shall be rejected.

4.2.3.3.2 *Velocity and pressure (for information only).* This information shall be obtained from the sample cartridges fired in the functioning test for the first three production lots. Results of this test will not be cause for rejection of any production lot except in the event of a premature burst. The results of this test shall be recorded (see 6.6).

4.2.3.3.3 *Premature bursts* (see 3.3.2) Critical defect — Code No. 10011. If any cartridge fails to comply with the applicable requirement during any test, the lot shall be rejected.

4.2.3.4 *Torque test* (see dwg. 9220299).

4.2.3.4.1 *Twenty-five- to thirty-inch pound disassembly torque — Critical Defect — Code No. 10012.* The cartridges shall be tested 100 percent for this requirement. Any cartridge which fails to meet the requirement shall be classed defective and removed from the lot or reworked to meet the requirement.

4.2.3.4.2 *Fifty inch-pound disassembly torque — Major defect — Code No. 10013.* A sample of fifty cartridges shall be tested for the requirement specified. If two or more cartridges fail the requirement the lot shall be rejected and reworked by loosening the cartridge container extension to the degree necessary to meet the fifty inch-pound disassembly torque.

4.2.4 *Inspection equipment.* Equipment Tabulation number ET–9220299 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with instructions in paragraph 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of

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stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and ~~maintenance~~ of the equipment listed on the ET shall be in accordance with MIL–I–45607.

4.2.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall, as required, make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verifying its accuracy shall be supplied by the contractor as required by the Government to perform its examination or test.

4.3 Test methods and procedures.

4.3.1 *Proving ground.*

4.3.1.1 *Functioning.* The cartridges shall be fired at a Government proving ground in accordance with the applicable acceptance test procedure. Observations shall be made for conformance with all applicable requirements.

4.3.1.2 *Refiring.* If for any reason the proof director considers that the test conditions have detrimentally affected the test results, he shall declare the results invalid and perform a new test on additional sample cartridges as required.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 *Level A.* The cartridges shall be packed and marked in accordance with dwg. 7549247 and dwg. 7549248.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number, and date of this specification.

(b) *Data cards.* Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL–STD–1167.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD — Contractor design on controlled contractor format and/or commercial equipment.

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MU — Army design, mandatory for use.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL–D–1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 *Submission of contractor designs.* Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND1. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Functioning. Satisfactory functioning shall be defined as functioning of the fuzes, ejection, ignition, and burning of the canister for the required time without detonation. (Detonation is disintegration of the canister assembly in the air after ejection from the projectile which results in a puff of smoke or dust cloud.

6.5 Combining of proving ground tests. When the contractor for the cartridge is the contractor for one or more of the components thereof, the proving ground tests of the components may be combined with the proving ground tests of the cartridge to save expense, upon agreement between the procuring activity and the contractor. In all cases where the cartridge specification does not cover all of the proving ground tests specified for the component, the addi-

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tional tests specified in the component specifications shall be conducted.

6.6 Submission of test data. In addition to the normal distribution of records when the cartridge is procured by the Department of the Army, one copy of all ballistic test data shall be forwarded to each of the following:

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA–ND7 and SMUPA–DG
Dover, New Jersey

6.7 Cost of deterioration check. The contracting officer shall arrange to have the contractor reimbursed for costs incurred in the performance of the check.

Custodian:

Army—MU

Preparing activity:

Army—MU

Project No. 1315–A549

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